UTM UNIVERSITI TEKNOLOGI MALAYSIA

SCHOOL OF COMPUTING
Faculty of Engineering

Semester II 2020/2021

Subject : Bioinformatics I (SCSB2103)
Section : 01 – Dr Haslina Hashim
Topic : Lab 03 – Pairwise Sequence Alignment

KOO XUAN (A23CS0300)

1) Perform pairwise alignment at the NCBI BLAST website.

i) Obtain the human HBA and HBB protein sequences.

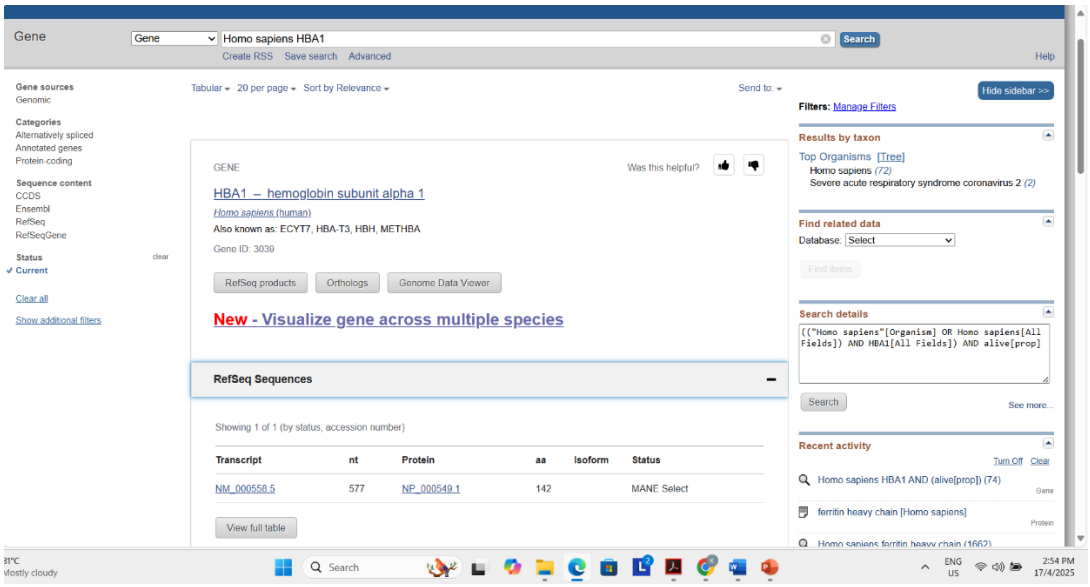
ii) Perform pairwise alignment at the NCBI BLAST website.

iii) Then use a comparison tool from the EBI website.

iv) Vary the scoring matrix (e.g., try different PAM and BLOSUM matrices) and record the effects on the score, the number of gaps, the percent identity, and the length of the aligned region.

v) For the NCBI BLASTP program note that the output of a pairwise alignment includes a dot matrix view.

HBA:

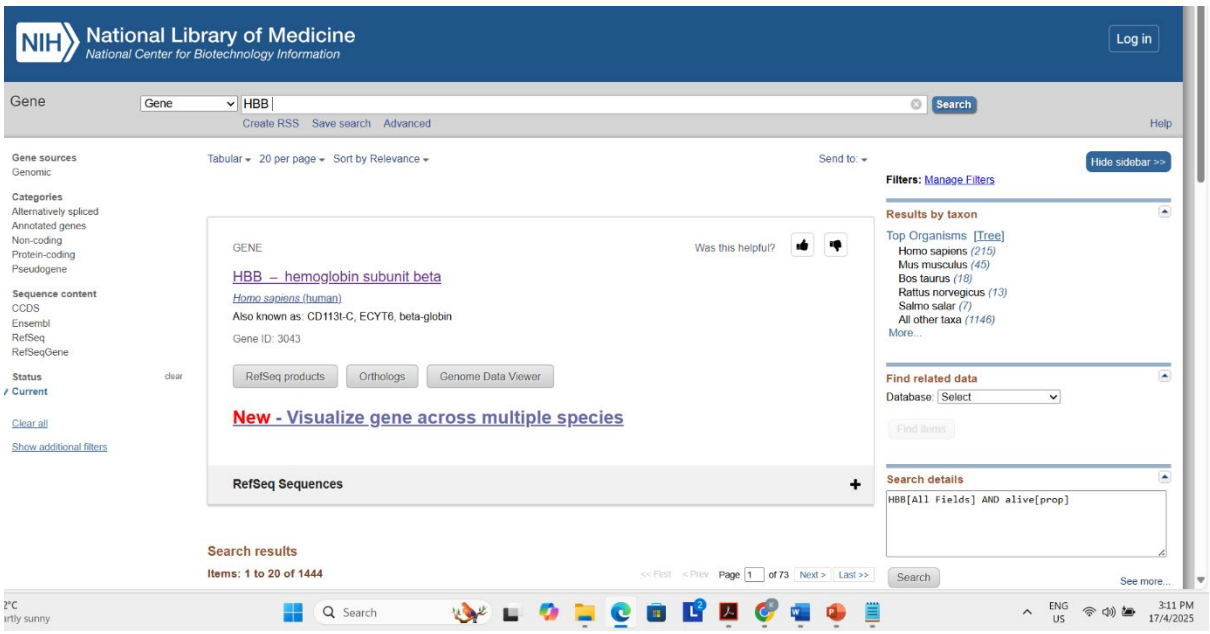


From here we can find the RefSeq accession number and sequence of HBA1 protein. They are:

>NP_000549.1 HBA1 [organism=Homo sapiens] [GeneID=3039]

MVLSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHFDLSHGSAQVKG
HGKKVADALTNAVAHVDDMPNALSALSDLHAHKLRVDPVNFKLLSHCLLVTLAAHLPA
EFTPAVHASLDKFLASVSTVLTSKYR

HBB:

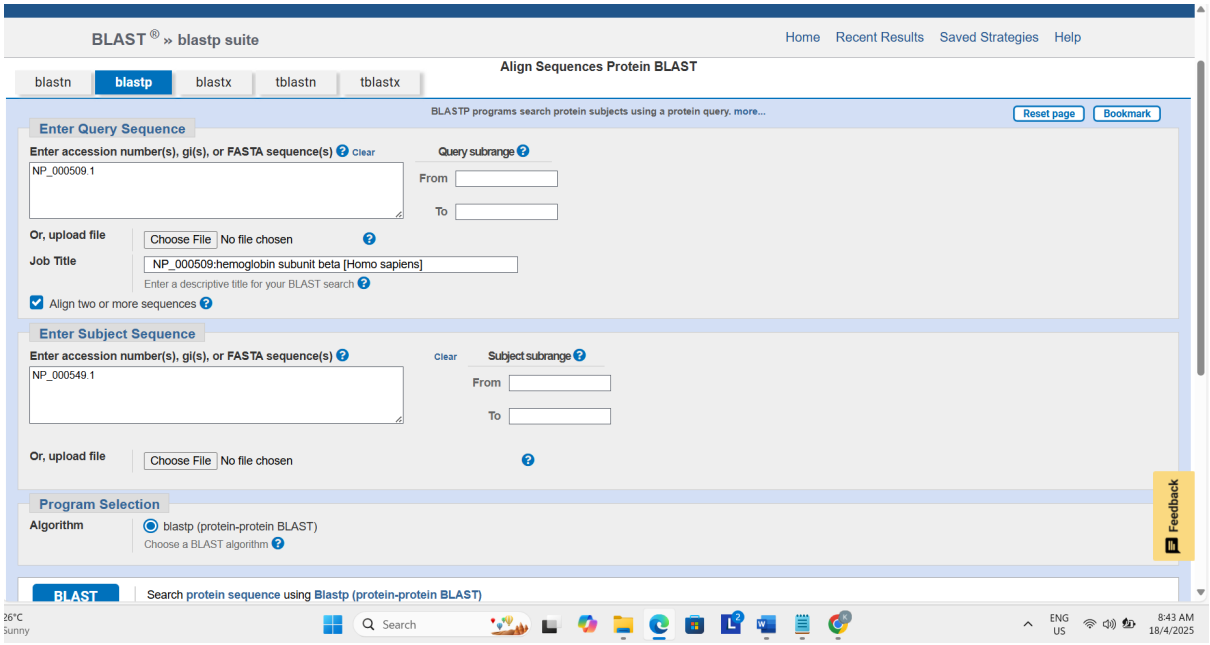


From here we can find the RefSeq accession number and sequence of HBB protein. They are:

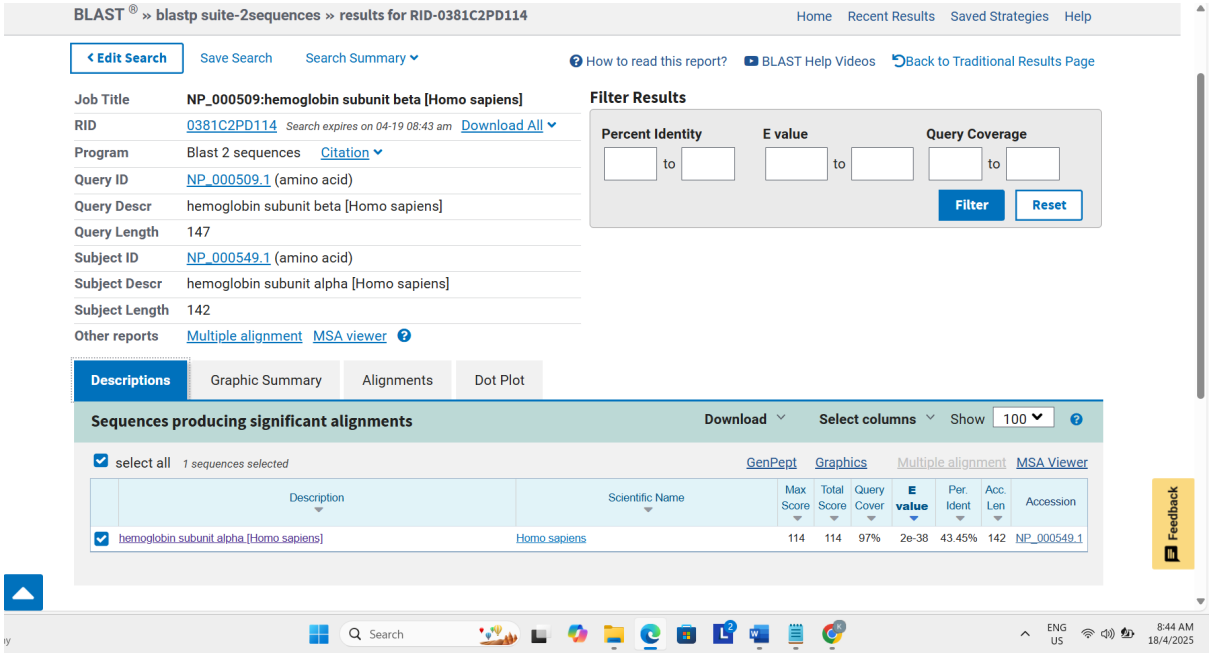
>NP_000509.1 HBB [organism=Homo sapiens] [GeneID=3043]

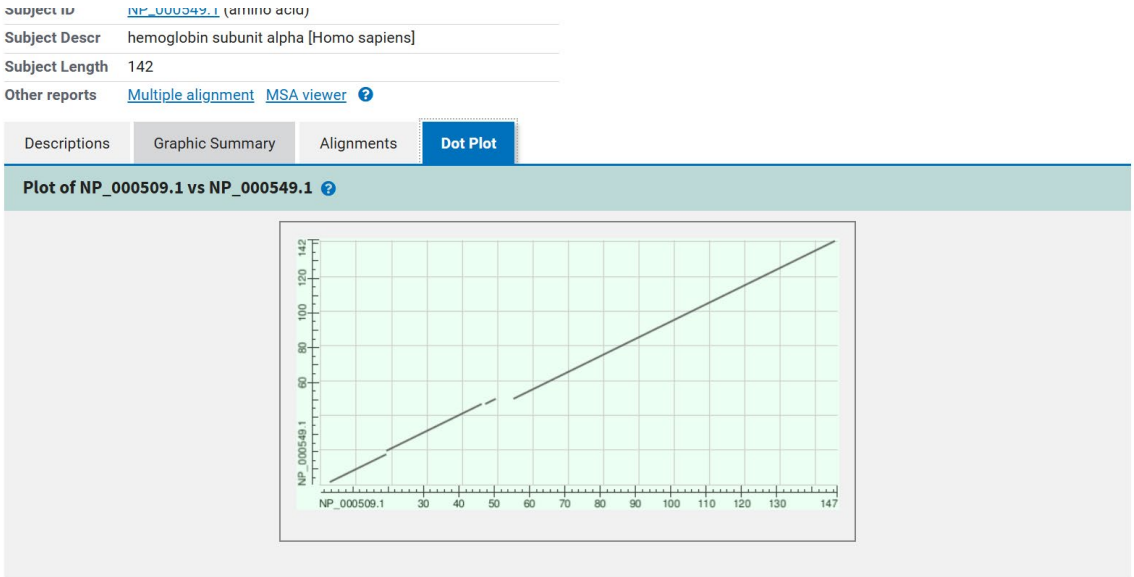
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGN
PKVKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLA
HHFGKEFTPPVQAAYQKVVAGVANALAHKYH

Then we align them at NCBI:

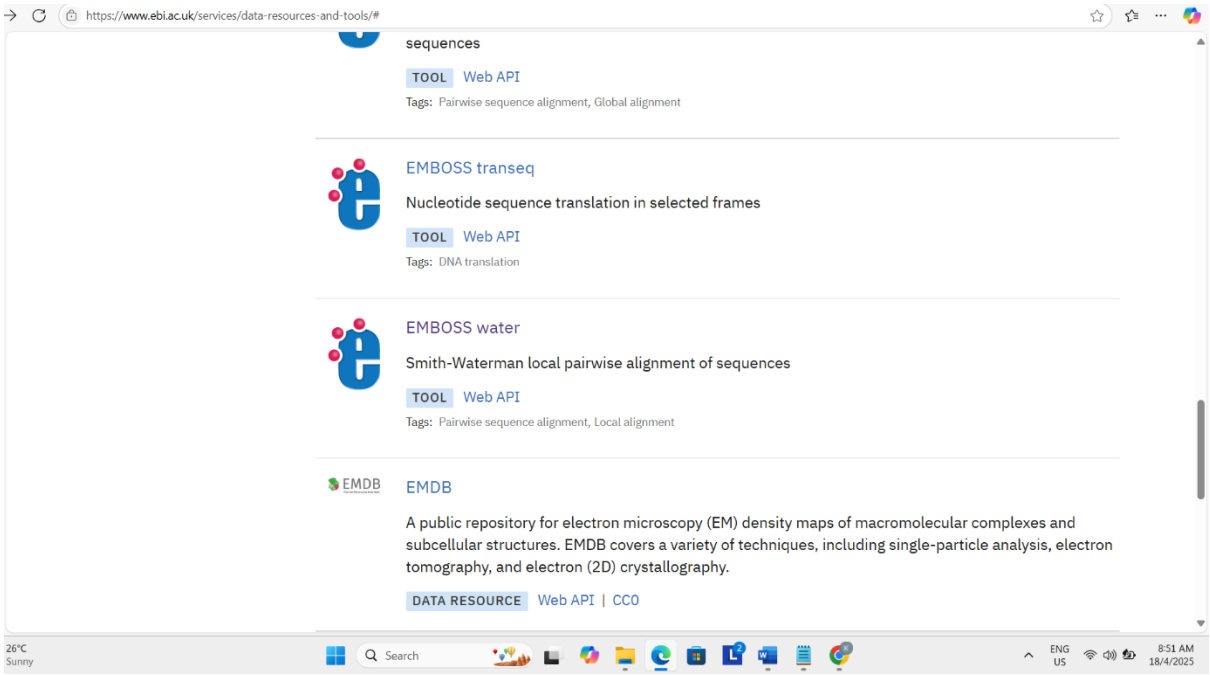


Here is part of the result, showing the description and the dot matrix view of the result:

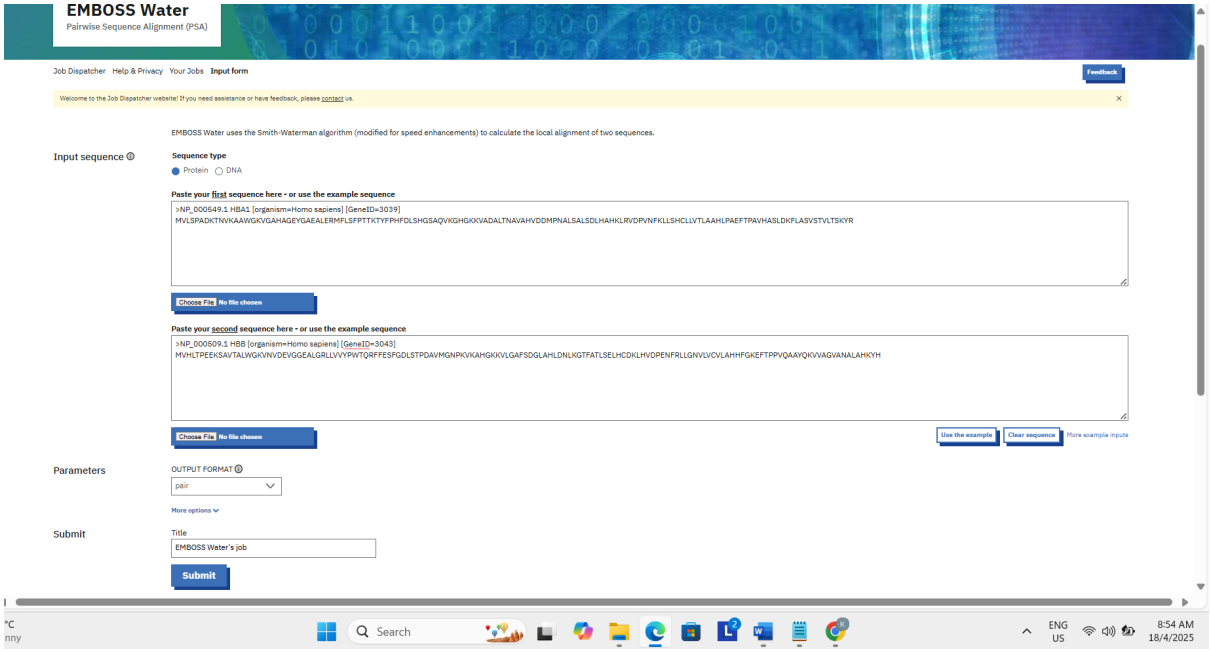




Visit EBI to try a search tool and see a list of relevant tools. Choose EMBOSS water.



Enter the two protein sequences in the FASTA format. Many tools do not accept RefSeq identifiers and therefore it is helpful to have the FASTA formatted sequences available.

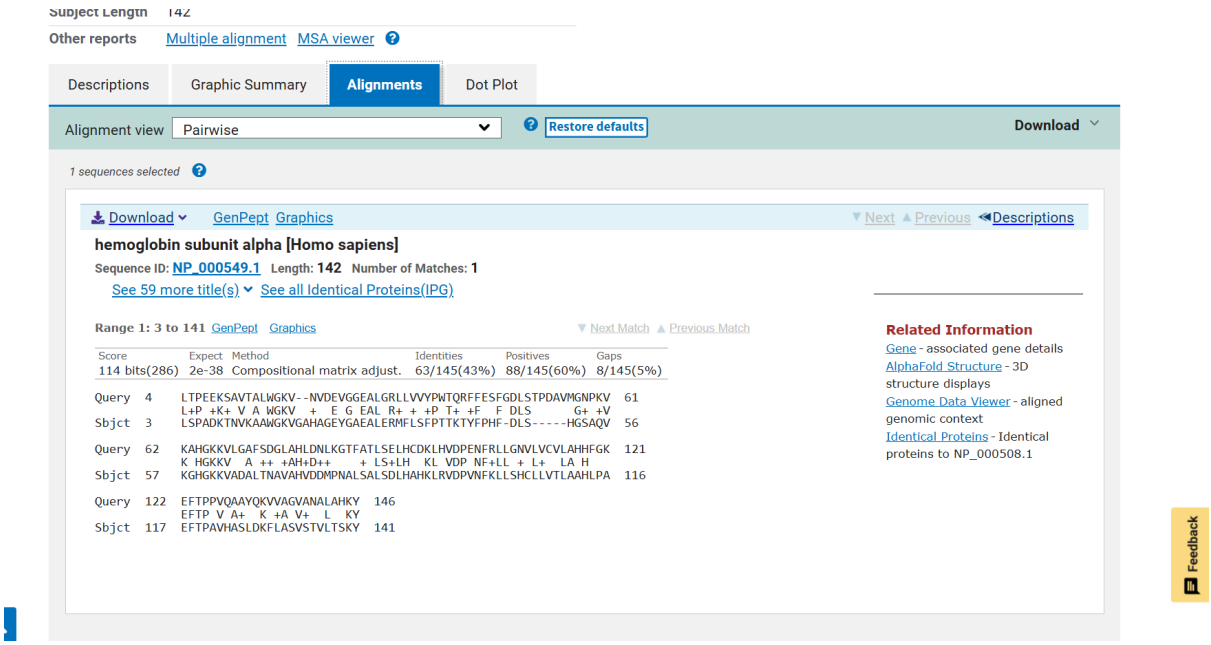


After submit, it results display:

```
#######################################
# Program: water
# Rundate: Fri 18 Apr 2025 01:54:35
# Commandline: water
#    -auto
#    -stdout
#    -asequence emboss_water-I20250418-015429-0554-62696334-p1m.asequence
#    -bsequence emboss_water-I20250418-015429-0554-62696334-p1m.bsequence
#    -datafile EBLOSUM62
#    -gapopen 10.0
#    -gapextend 0.5
#    -aformat3 pair
#    -sprotein1
#    -sprotein2
# Align_format: pair
# Report_file: stdout
#######################################

#=======================================
#
# Aligned_sequences: 2
# 1: NP_000549.1
# 2: NP_000509.1
# Matrix: EBLOSUM62
# Gap_penalty: 10.0
# Extend_penalty: 0.5
#
# Length: 145
# Identity:      63/145 (43.4%)
# Similarity:    88/145 (60.7%)
# Gaps:           8/145 ( 5.5%)
# Score: 293.5
#
#
#=======================================
```

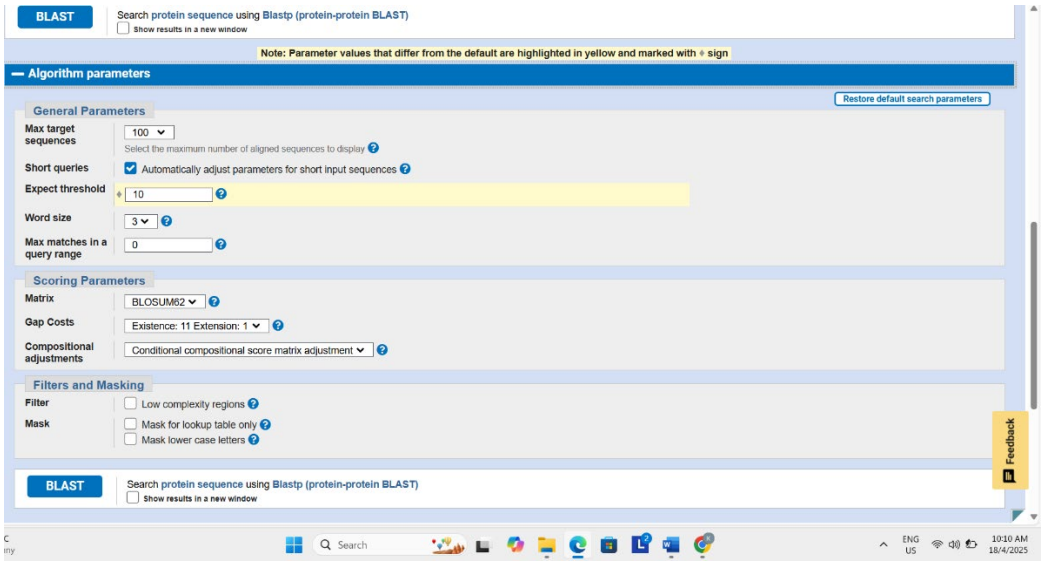
```
NP_000549.1        3 LSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHF-DLS-     50
                     |:|.:|:.|.|.||||  :..|.|.|||.|:.:.:|.|:.:|..| |||
NP_000509.1        4 LTPEEKSAVTALWGKV--NVDEVGGEALGRLLVVYPWTQRFFESFGDLST     51

NP_000549.1       51 ----HGSAQVKGHGKKVADALTNAVAHVDDMPNALSALSDLHAHKLRVDP     96
                         .|:.:||.|||||..|.::.:||:|::....:.||:||..||.|||
NP_000509.1       52 PDAVMGNPKVKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDP    101

NP_000549.1       97 VNFKLLSHCLLVTLAAHLPAEFTPAVHASLDKFLASVSTVLTSKY    141
                     .||:||.:.|:..||.|...||||.|.|:..|.:|.|:..|..||
NP_000509.1      102 ENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKY    146


#---------------------------------------
#---------------------------------------
```

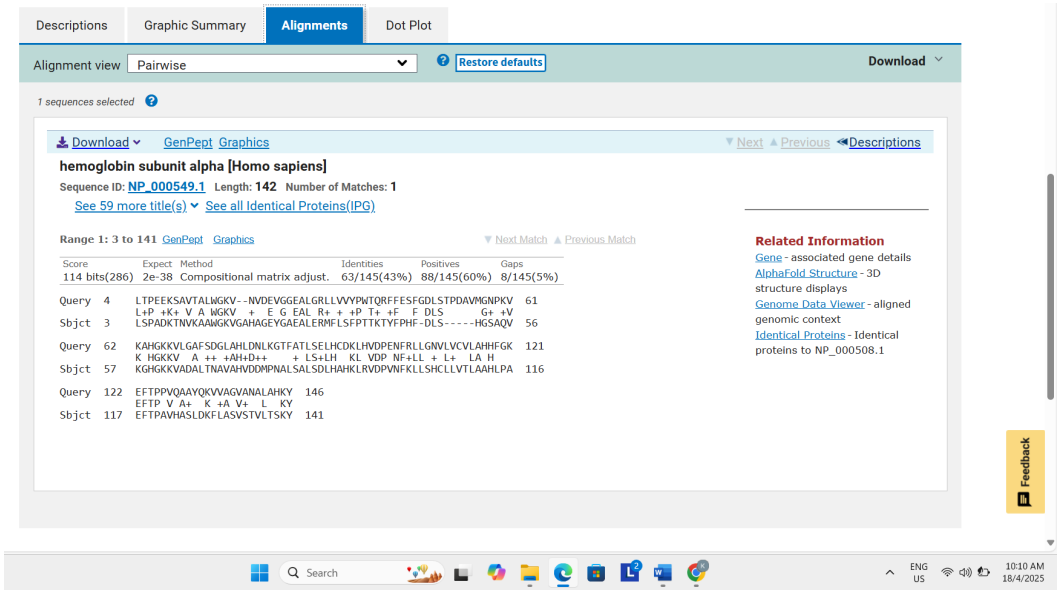
Then, return to the NCBI. It was:



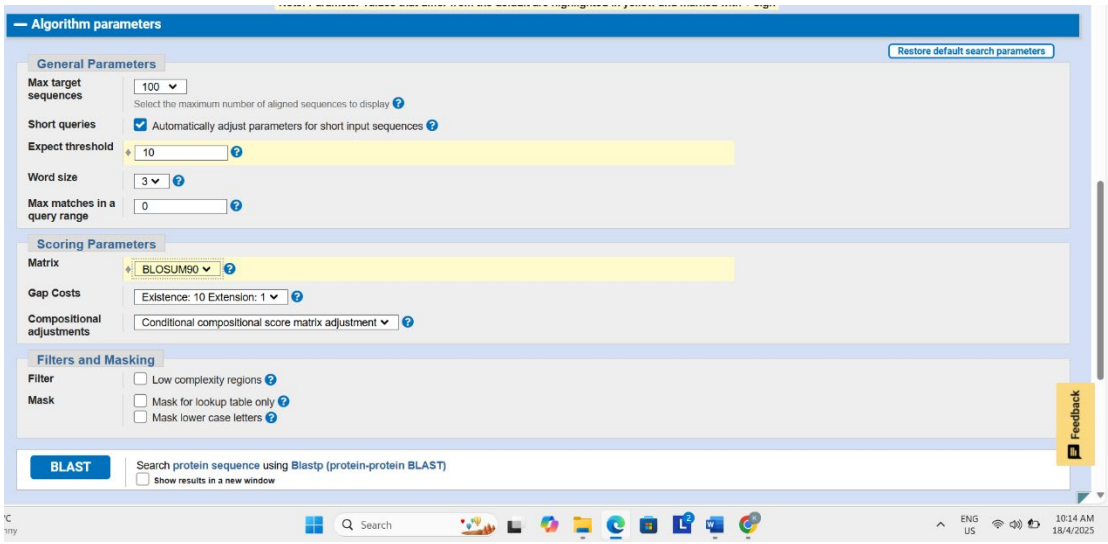
Next, click “edit and resubmit” at the top left of the BLAST result page to perform pairwise alignment again. Click “algorithm parameters” to change those settings:

BLOSUM62





BLOSUM90



Alignment view Pairwise Restore defaults Download

1 sequences selected

Download GenPept Graphics Next Previous Descriptions

hemoglobin subunit alpha [Homo sapiens]

Sequence ID: NP_000549.1 Length: 142 Number of Matches: 1

See 59 more title(s) See all Identical Proteins(IPG)

Range 1: 3 to 141 GenPept Graphics Next Match Previous Match

Score	Expect	Method	Identities	Positives	Gaps
118 bits(275)	4e-37	Compositional matrix adjust.	63/145(43%)	81/145(55%)	8/145(5%)

```
Query  4    LTPEEKSAVTALWGKV--NVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPKV  61
            L+P +K+ V A WGKV     E G EAL R+   +P T+ +F  F DLS      G  +V
Sbjct  3    LSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHF-DLSH-----GSAQV  56

Query  62   KAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFGK  121
            K HGKKV  A ++  AH D++    + LS+LH  KL VDP NF+LL   L   LA H
Sbjct  57   KGHGKKVADALTNAVAHVDDMPNALSALSDLHAHKLRVDPVNFKLLSHCLLVTLAAHLPA  116

Query  122  EFTPPVQAAYQKVVAGVANALAHKY  146
            EFTP V+A+  K  A V+  L  KY
Sbjct  117  EFTPAVHASLDKFLASVSTVLTSKY  141
```

Related Information

Gene - associated gene details
AlphaFold Structure - 3D structure displays
Genome Data Viewer - aligned genomic context
Identical Proteins - Identical proteins to NP_000508.1

PAM30

Restore default search parameters

General Parameters

Max target sequences: 100
Select the maximum number of aligned sequences to display

Short queries: Automatically adjust parameters for short input sequences

Expect threshold: 10

Word size: 3

Max matches in a query range: 0

Scoring Parameters

Matrix: PAM30

Gap Costs: Existence: 9 Extension: 1

Compositional adjustments: Conditional compositional score matrix adjustment

Filters and Masking

Filter: Low complexity regions

Mask: Mask for lookup table only; Mask lower case letters

BLAST Search protein sequence using Blastp (protein-protein BLAST)
Show results in a new window

Descriptions Graphic Summary **Alignments** Dot Plot

Alignment view Pairwise Restore defaults Download

1 sequences selected

Download GenPept Graphics Next Previous Descriptions

hemoglobin subunit alpha [Homo sapiens]

Sequence ID: NP_000549.1 Length: 142 Number of Matches: 1

See 59 more title(s) See all Identical Proteins(IPG)

Range 1: 1 to 142 GenPept Graphics Next Match Previous Match

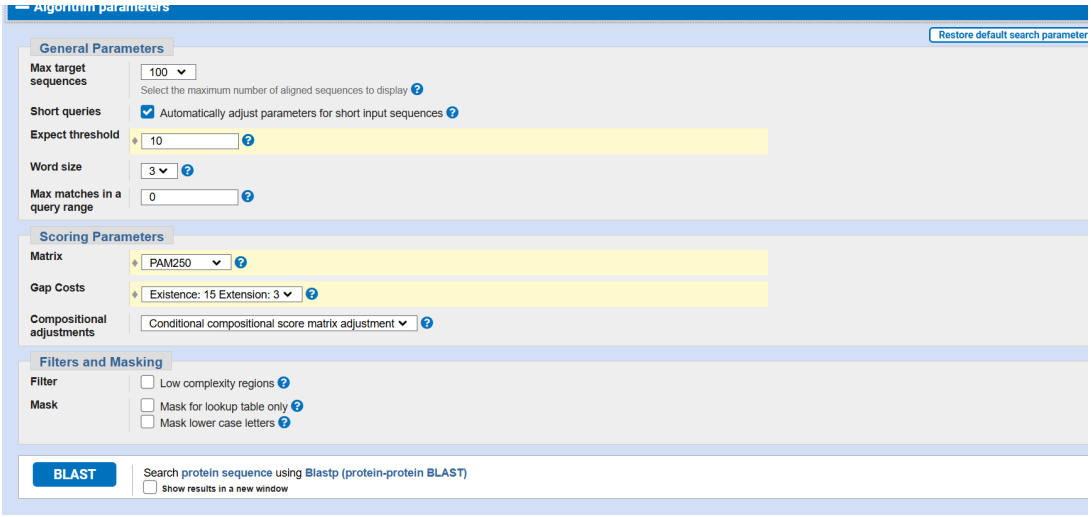
Score	Expect	Method	Identities	Positives	Gaps
121 bits(278)	4e-34	Compositional matrix adjust.	65/149(44%)	74/149(49%)	9/149(6%)

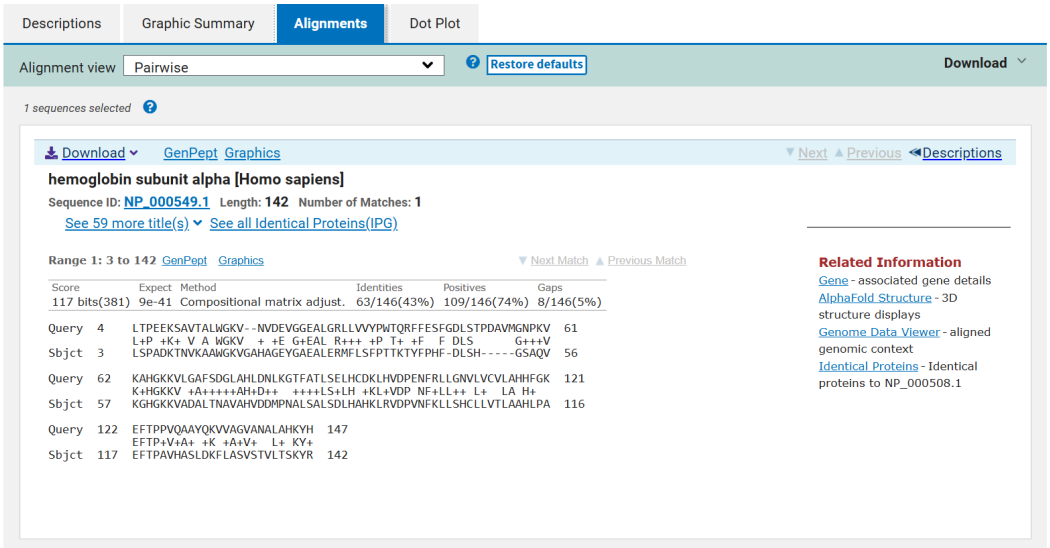
```
Query  1    MVHLTPEEKSAVTALWGKV--NVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGN  58
            MV L P +K  V A WGKV     E G EAL R+   +P T  +F  F DLS      G
Sbjct  1    MV-LSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHF-DLSH-----GS  53

Query  59   PKVKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHH  118
             VK HGKKV  A  +  AH D++      LS+LH  KL VDP NF LL   L   LA H
Sbjct  54   AQVKGHGKKVADALTNAVAHVDDMPNALSALSDLHAHKLRVDPVNFKLLSHCLLVTLAAH  113

Query  119  FGKEFTPPVQAAYQKVVAGVANALAHKYH  147
             EFTP V+A   K  A V   L  KY
Sbjct  114  LPAEFTPAVHASLDKFLASVSTVLTSKYR  142
```

Related Information

Gene - associated gene details
AlphaFold Structure - 3D structure displays
Genome Data Viewer - aligned genomic context
Identical Proteins - Identical proteins to NP_000508.1

PAM250





Settings	Matrix	E value	Identities	Gaps
Default (see above)	BLOSUM62	2e-38	63/145	8/145
	BLOSUM90	4e-37	63/145	8/145
	PAM30	4e-34	65/149	9/149
	PAM250	9e-41	63/146	8/146

Using these other scoring matrices has a minimal effect on the results. It is often easier to see large differences when two proteins are more distantly related, or when only a portion of two proteins aligns well.

2) Perform pairwise alignment at the UCSC website.

i) Go to http://genome.ucsc.edu. Follow the link to the genome browser, select the human genome hg19 build, and enter a query of hbb. This should direct you to chr11:5,246,696-5,248,301 (a region of 1,606 base pairs encompassing the beta globin gene, HBB.

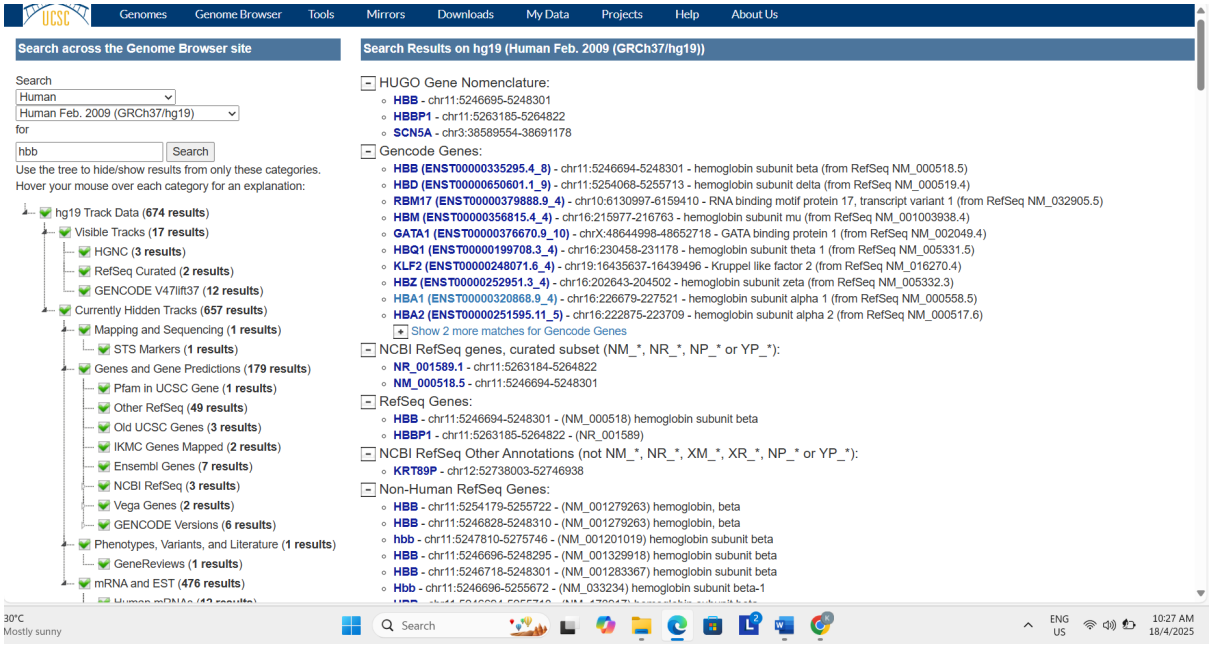
ii) Click the box to set the view to default tracks.

iii) Under Comparative Genomics" select Placental Chain/Net and set the display to full. By clicking the Placental Chain/Net header you can view a series of options. Set Chains to full view and Nets to full view. Set the species to horse (deselect other species). Click submit.

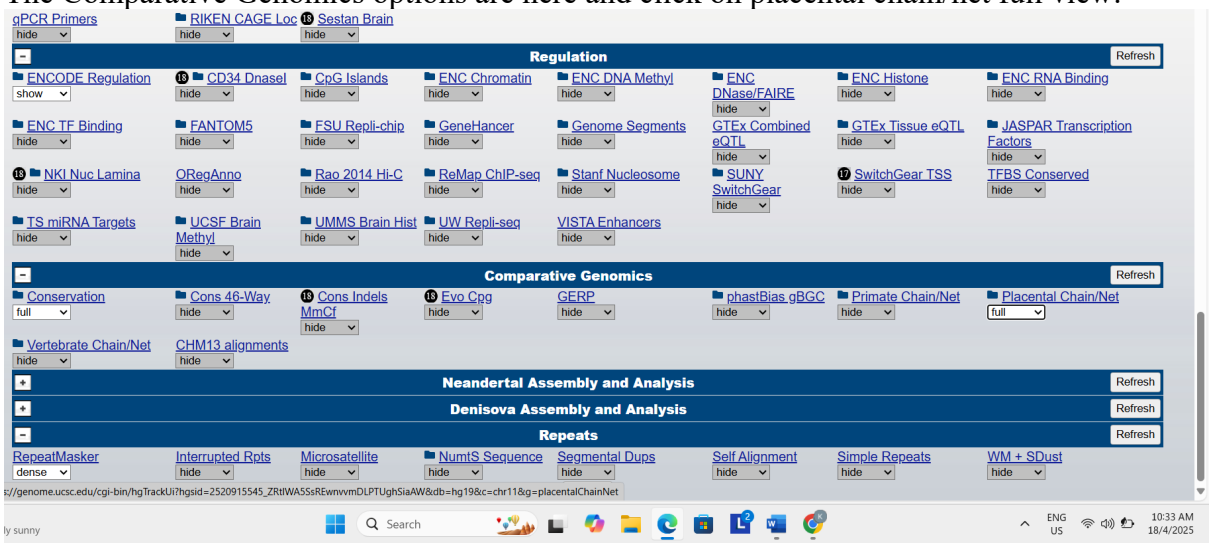
iv) The display now shows human/horse chained alignments and and alignment nets.

Enter hbb at UCSC Genome Browser

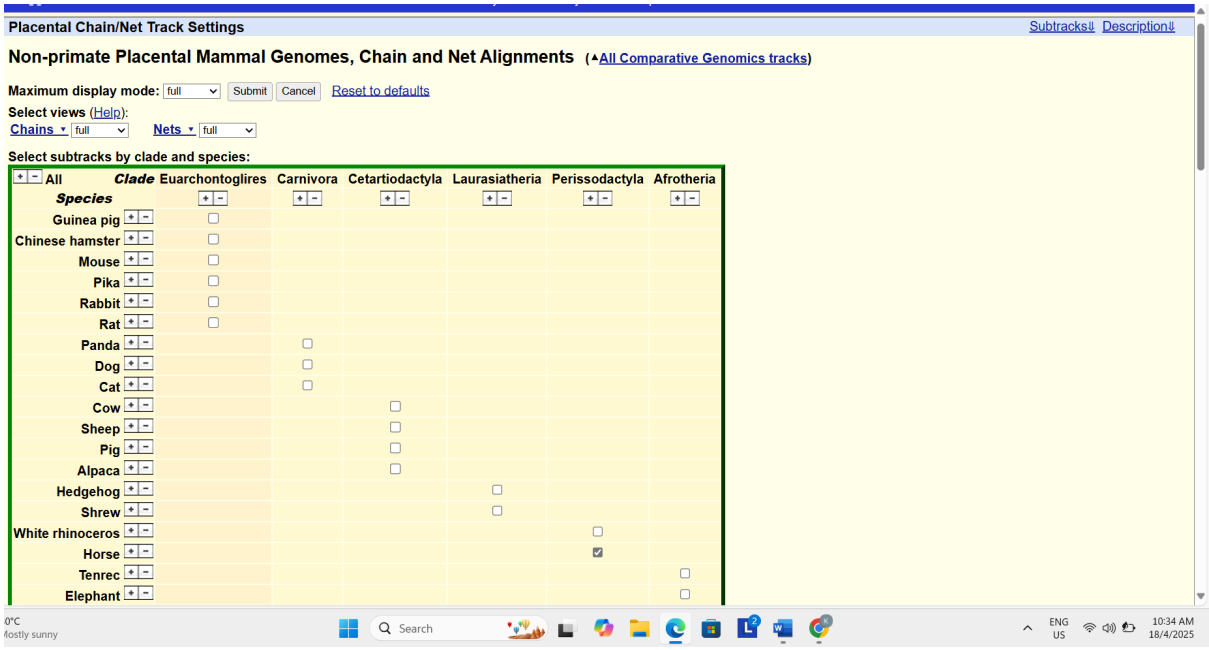
Then click HBB:



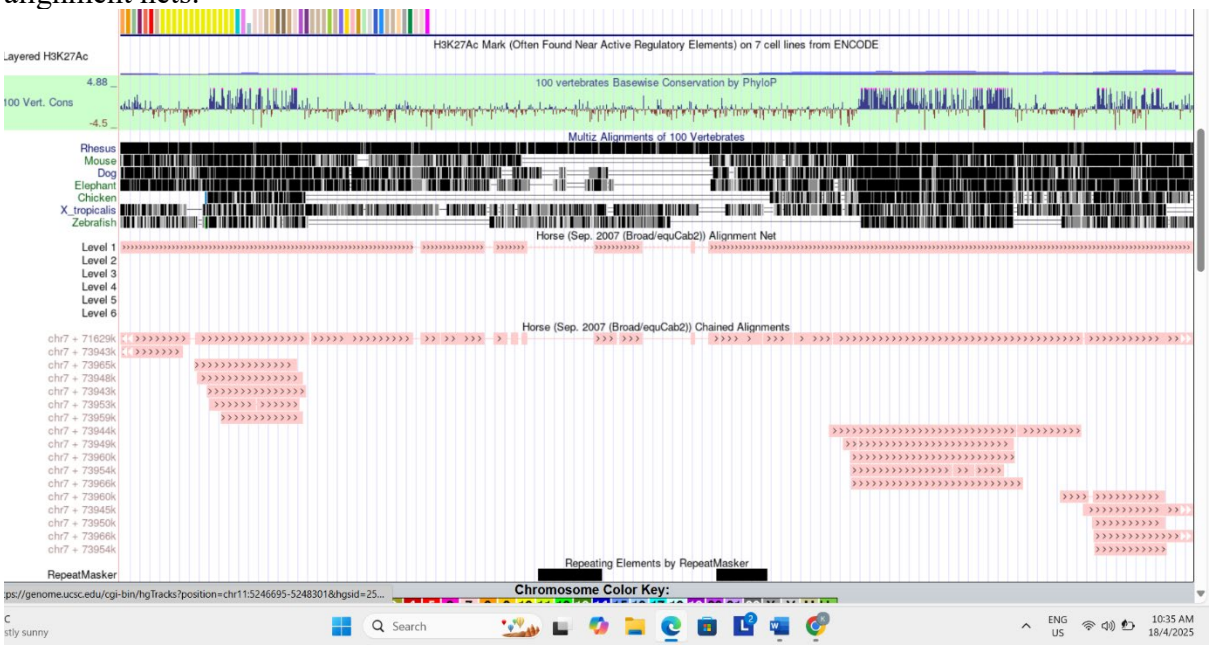
The Comparative Genomics options are here and click on placental chain/net full view:



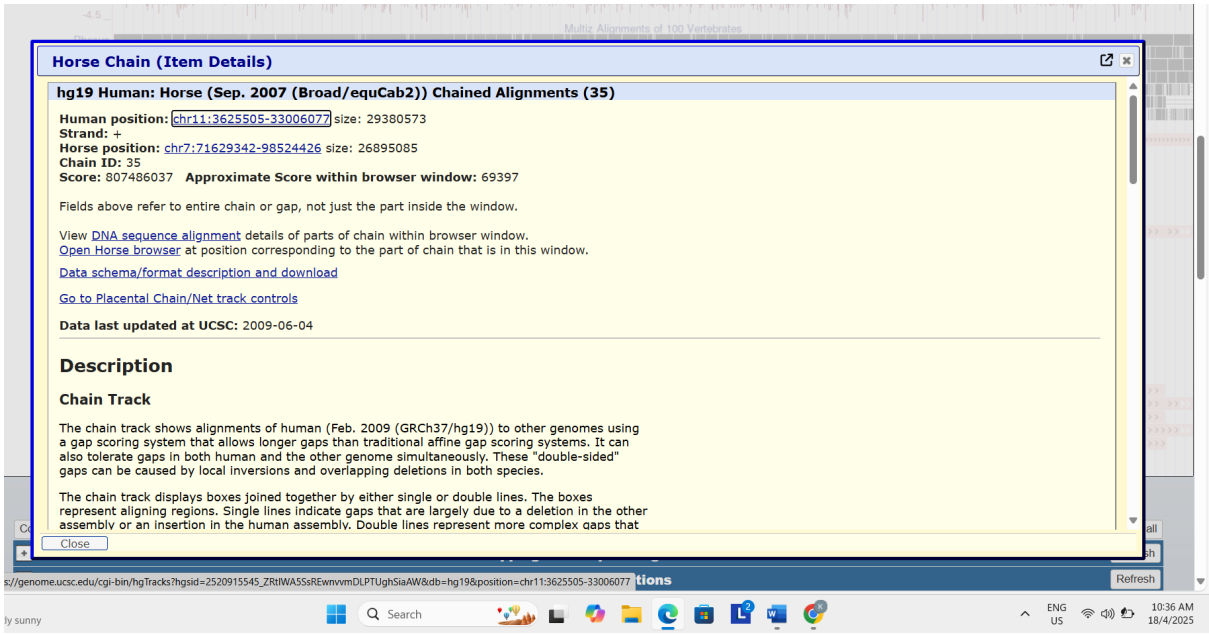
Click the header of that track to view more options; deselect all (-key) then select horse, and set chains and nets from pack or dense to full:



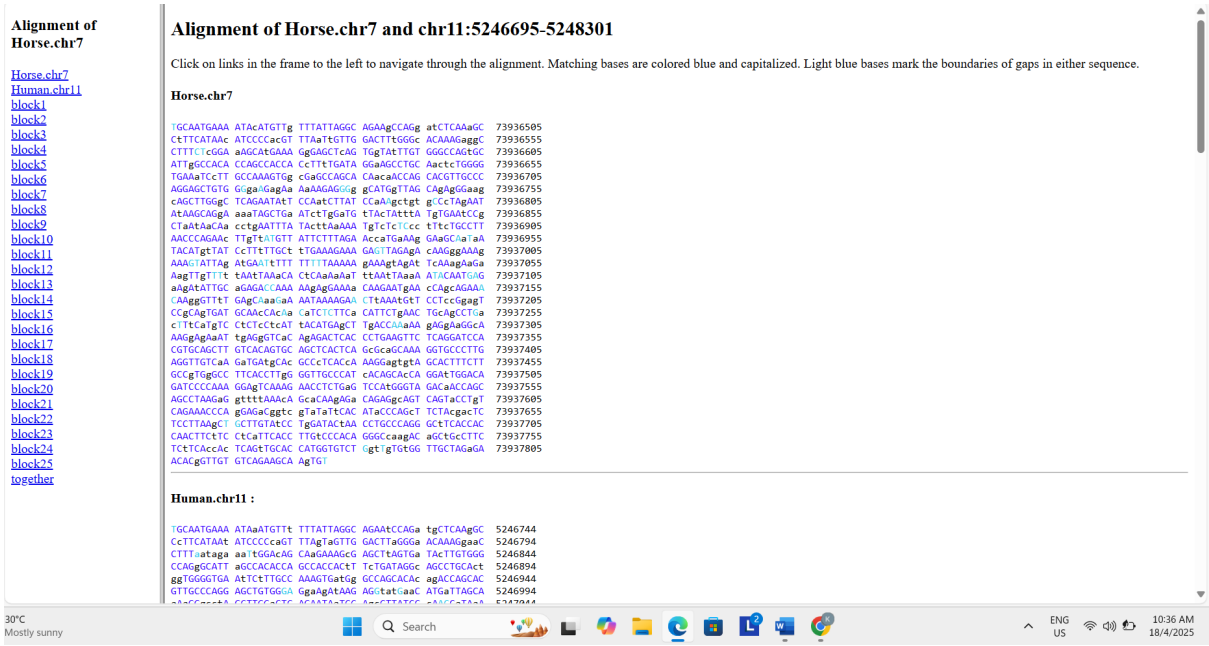
Click Submit. The resulting Genome Browser display shows chained alignments and alignment nets.



Click an alignment block to to view the pairwise alignment between human and horse DNA at that region.



The results shows here:



3) Using the amino acid explorer tool from NCBI

i) Visit http://www.ncbi.nlm.nih.gov/Class/Structure/aa/aa_explorer.cgi Select the Biochemical Properties table. Which amino acid is most abundant? (Is it leucine, at 9.94%?). Use this table to test yourself and make sure you know the one- and three-letter abbreviations for all 20 amino acids, as well as their structures.

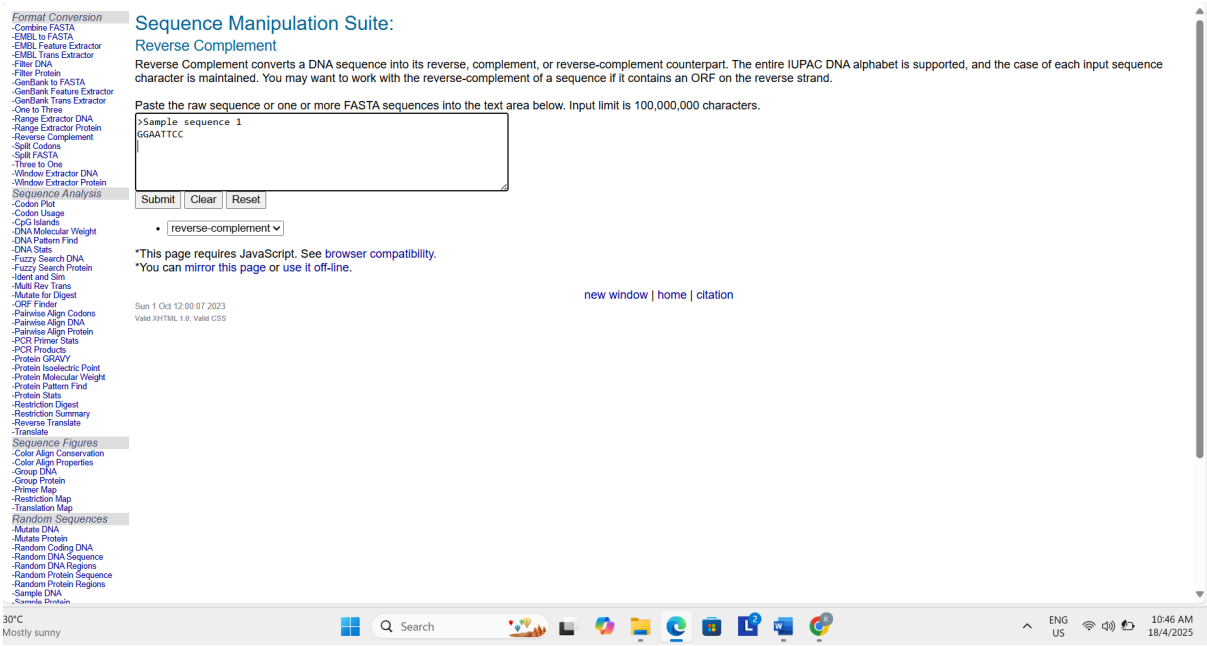
ii) Is tyrosine a hydrophobic amino acid? To decide, use the Common Substitutions table. Explore valine (a hydrophobic residue), sort the results by hydrophobicity, and see where tyrosine is located. You can also explore the Structure and Chemistry table.

The website doesn't exits

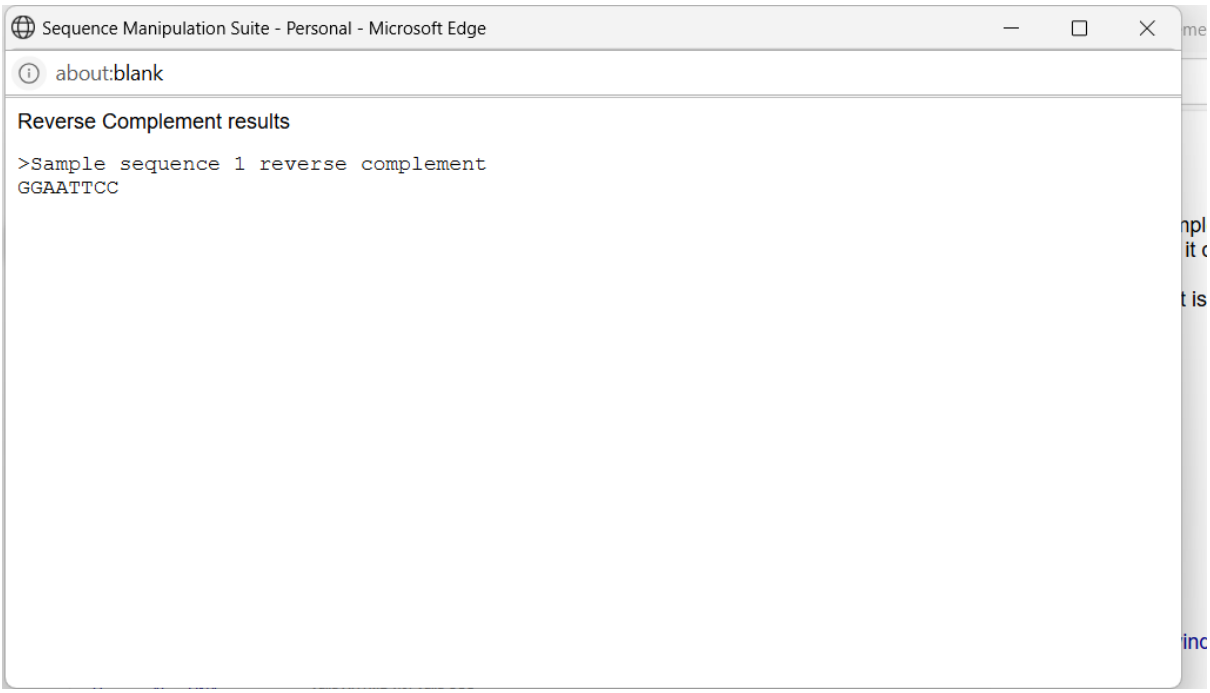
4) Sequence Manipulation Suite

i) Many tools are available to manipulate sequences. Visit the Sequence Manipulation Suite (http://www.bioinformatics.org/sms2/index.html) to access a large number of tools. What is the reverse complement of the sequence GGAATTCC?

Visit website http://www.bioinformatics.org/sms2/index.html , then click reverse complement and enter sequence GGAATTCC in the enter box and click submit.



The results show:



5) Perform pairwise alignment using EMBOSS tools via Galaxy and UCSC

i) In this exercise we will perform global alignment with the EMBOSS package needle and local alignment with the EMBOSS package water.

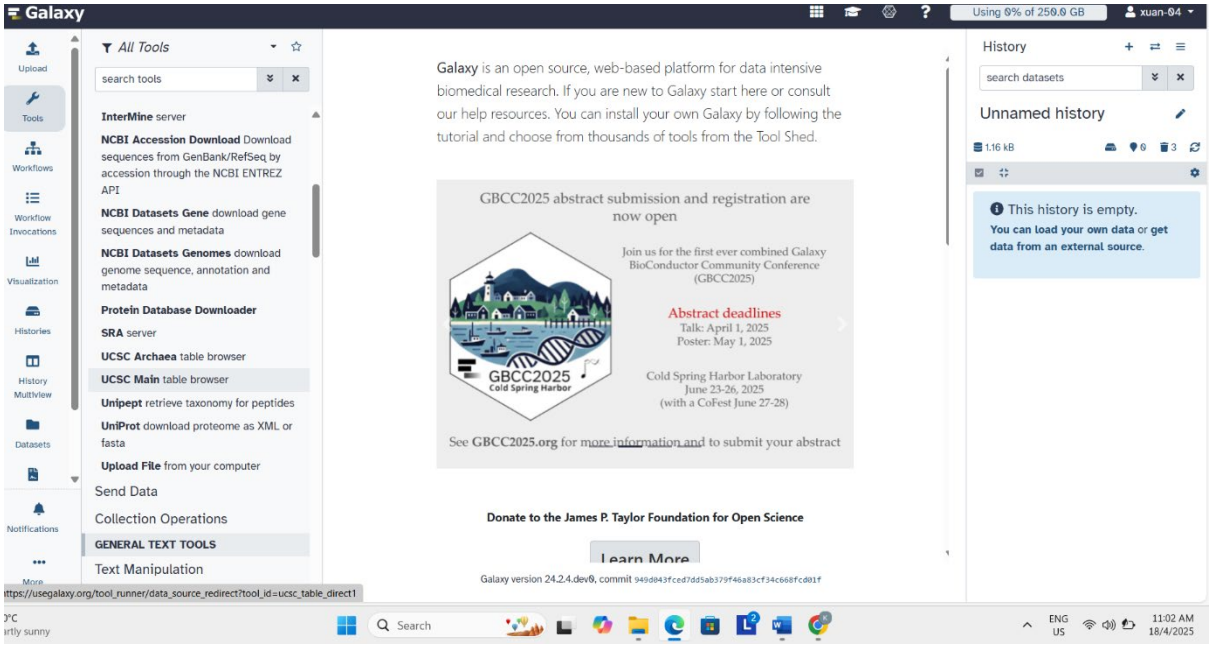
ii) Both of these are available at the Galaxy public web server (along with over 100 other EMBOSS tools). Box 3.9 introduces EMBOSS and explains how to import beta globin (HBB) and alpha globin (HBA2) proteins from the UCSC Table Browser using Galaxy, and to then align them. This history is saved at https://main.g2.bx.psu.edu/u/pevsner/h/pairwise-alignment-via-ucsc-and-emboss (WebLink 3.14).

iii) Note that Galaxy is a web-based platform for using hundreds of bioinformatics tools, including next-generation sequence data analysis software.

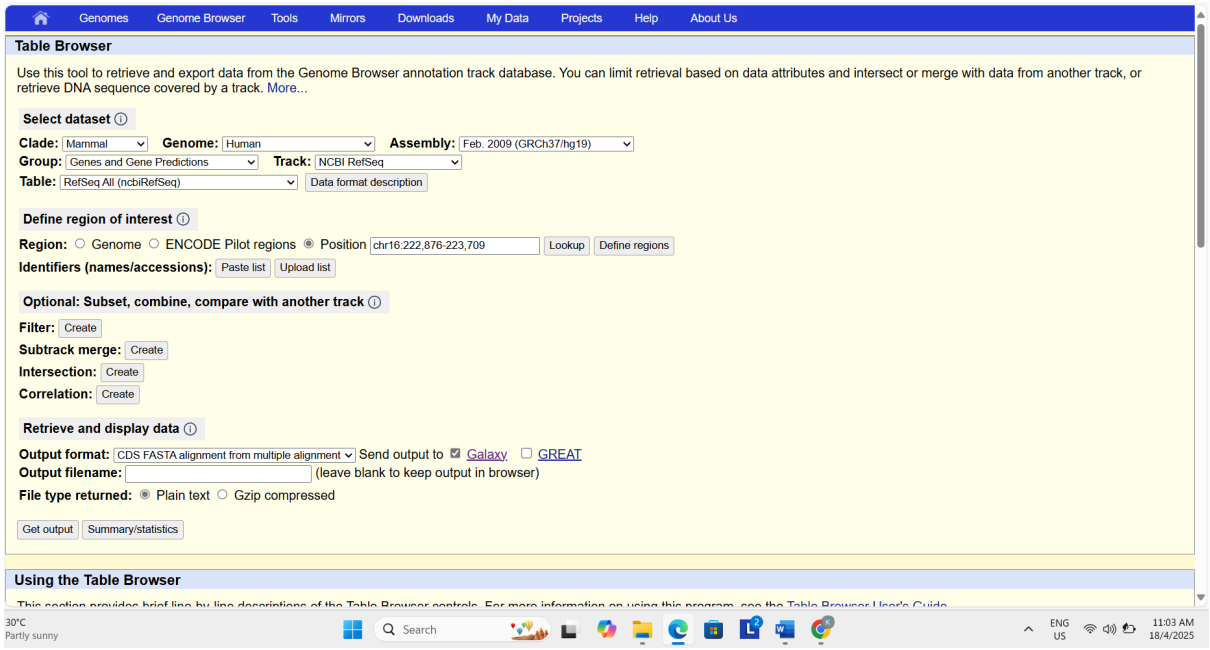
iv) To use it visit http://usegalaxy.org then go to the public server.

v) Be sure to create a username and log in. This will allow you to continue your work over time and at different work stations.

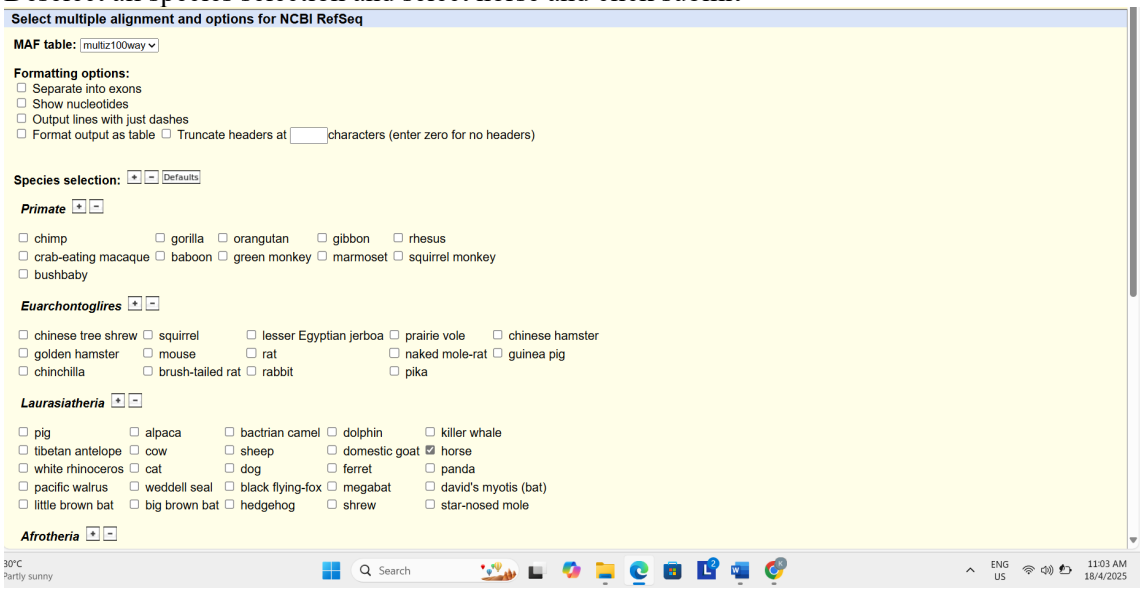
First, visit galaxy website and log in, and then go to UCSC Main table browser



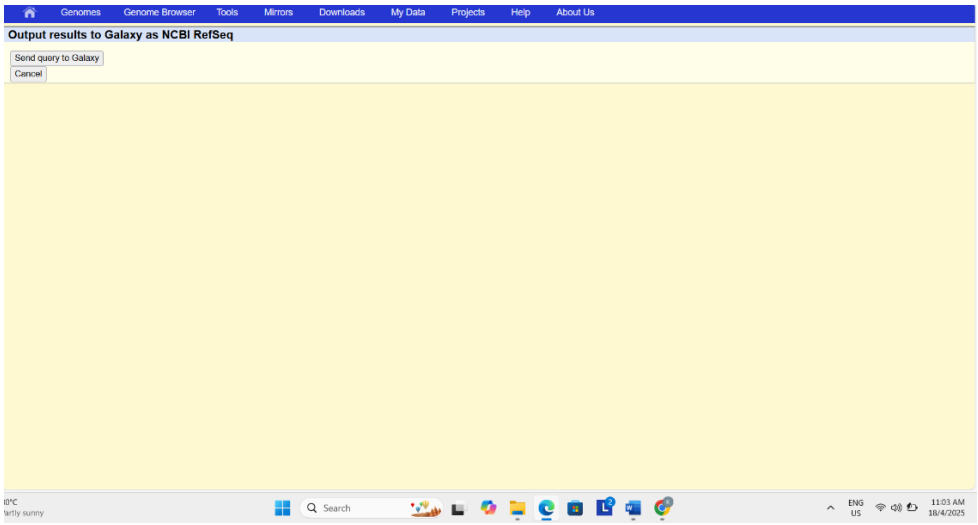
Then enter HBA2 at the position and select FASTA output format and get output



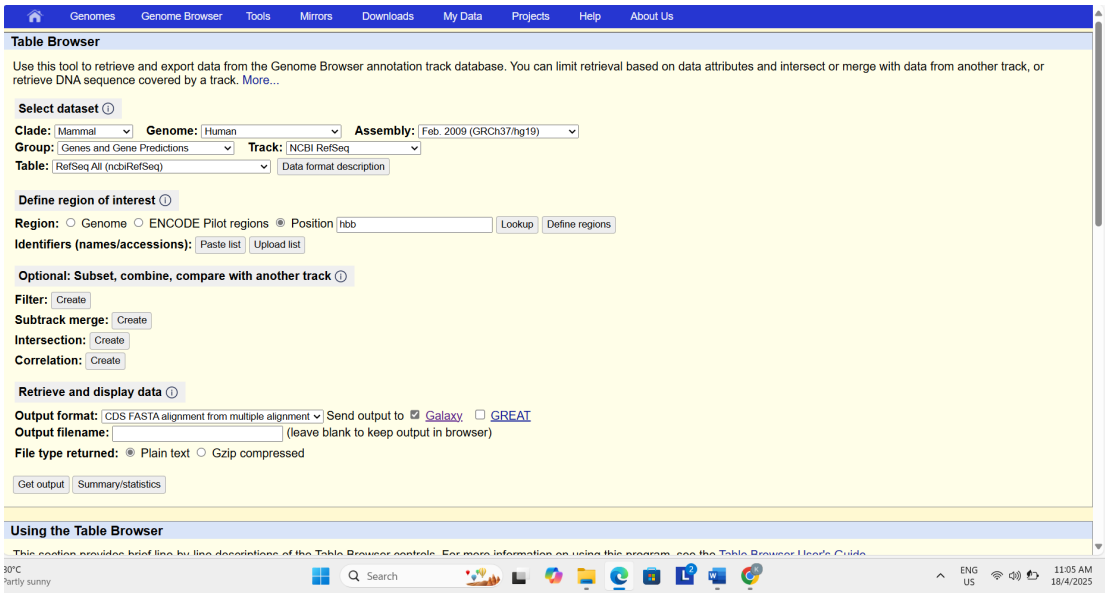
Deselect all species selection and select horse and click submit

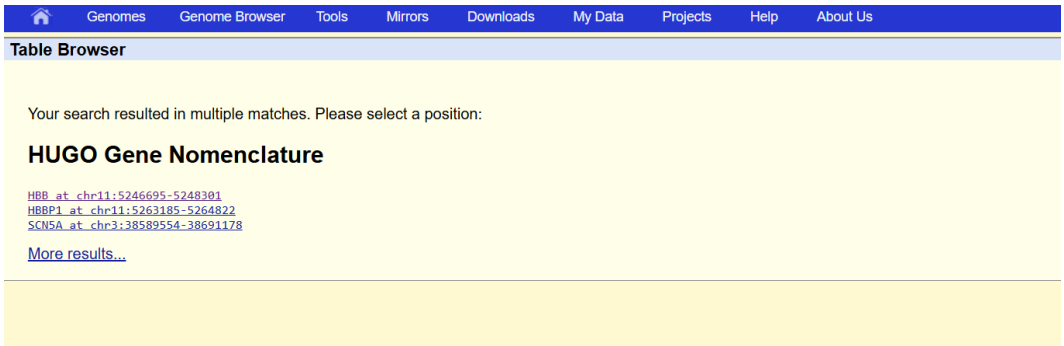


Click send query to galaxy

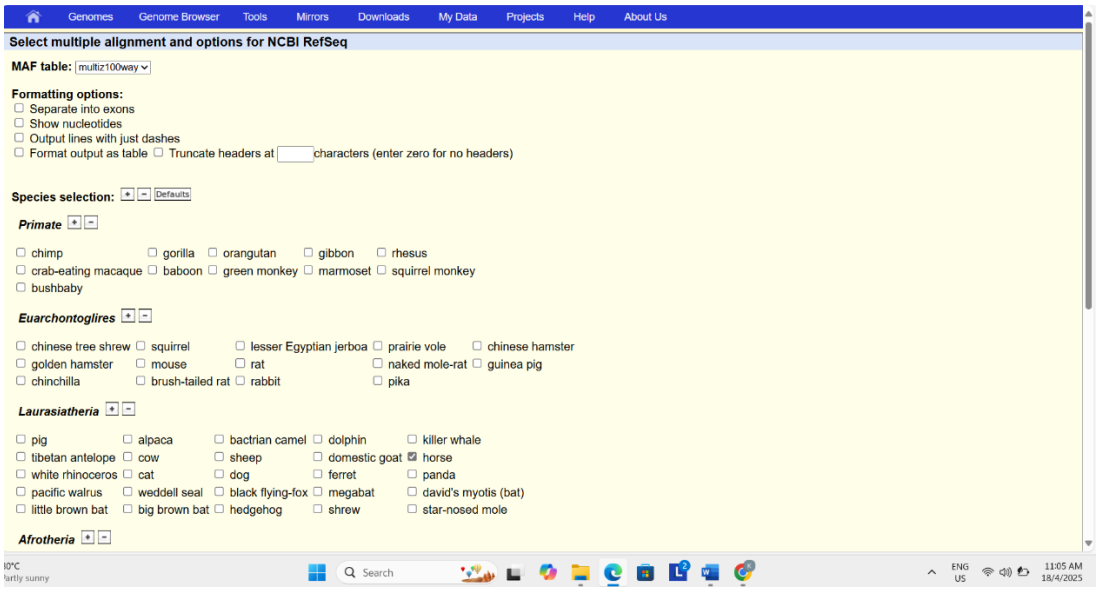


Repeat again and put HBB for the position:



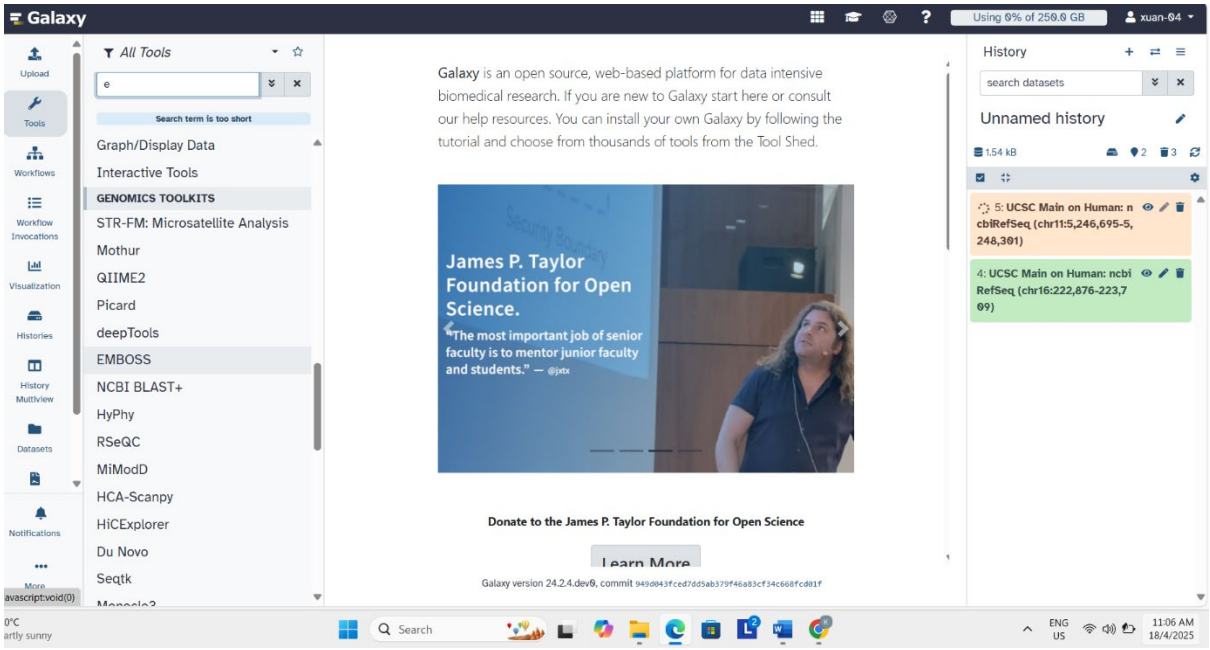


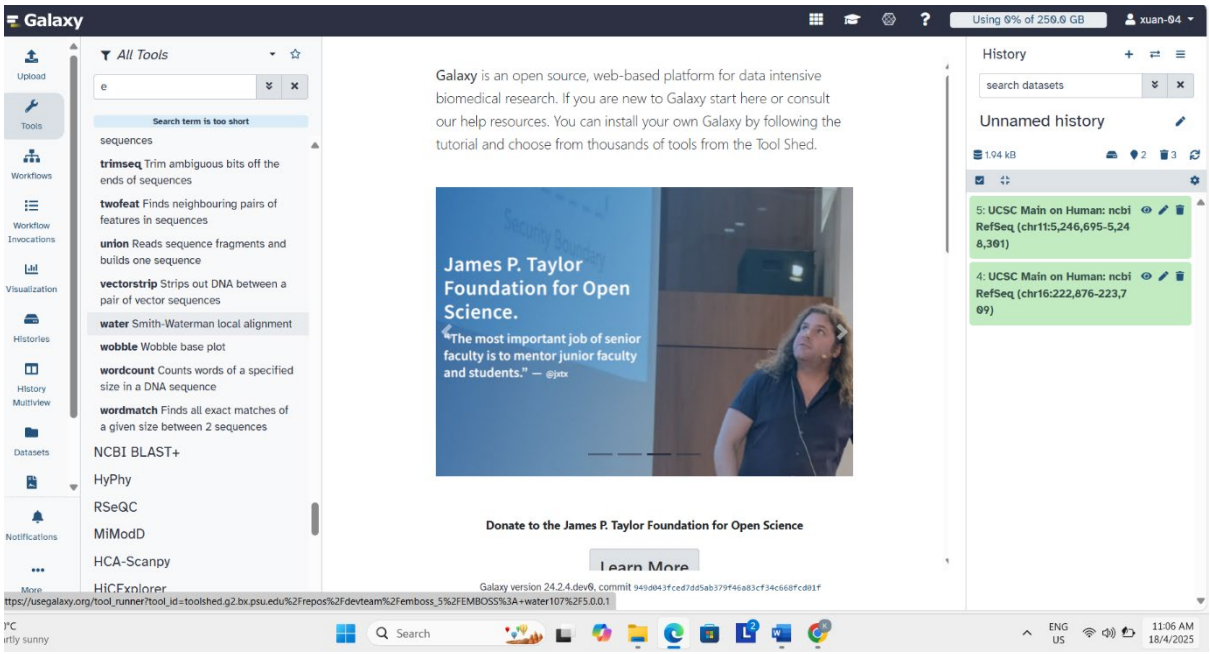
Deselect all species selection and select horse and click submit



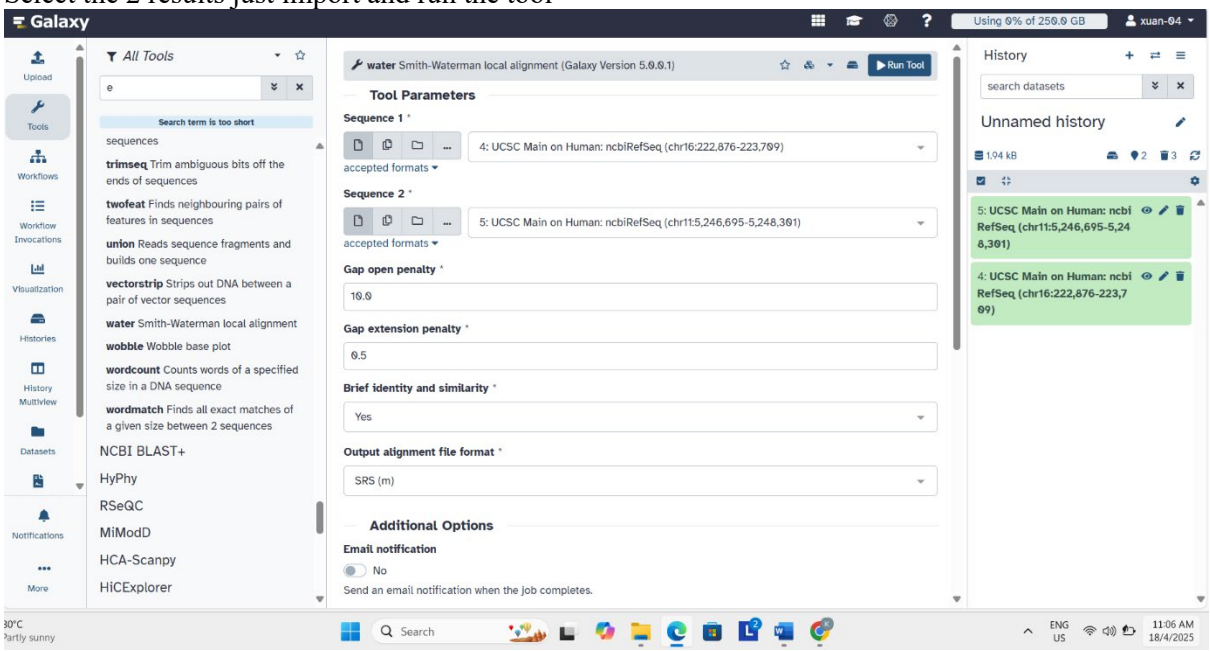
Click send query to galaxy

After the query successfully add to galaxy website, search for EMBOSS water

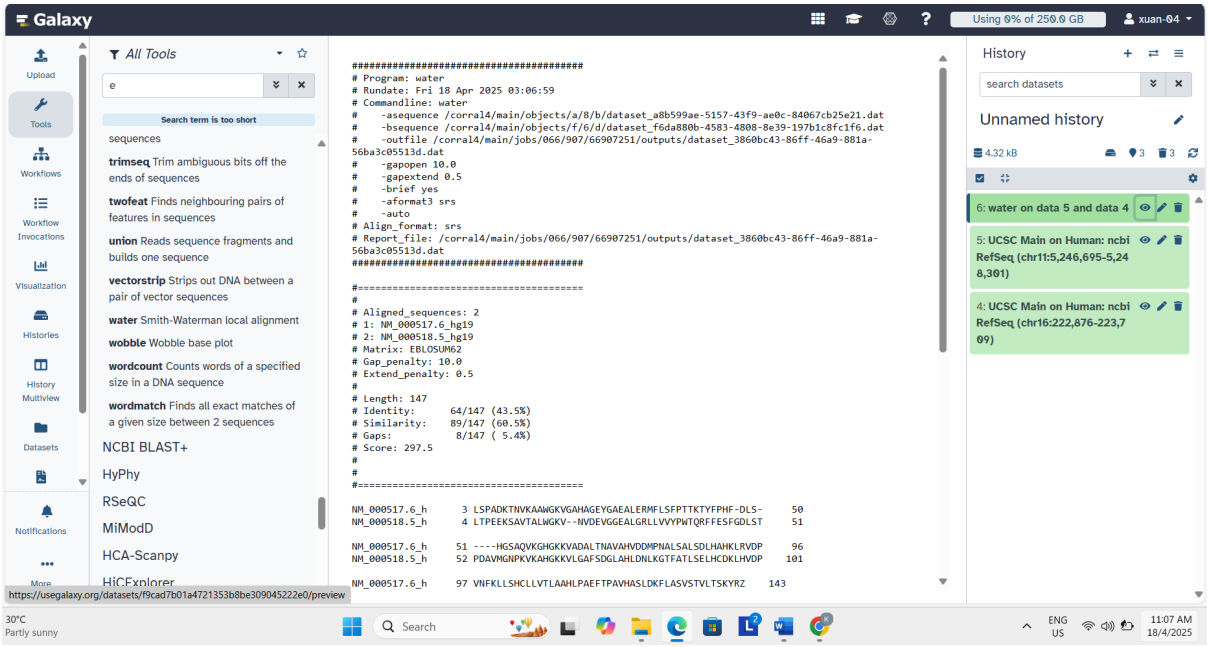




Select the 2 results just import and run the tool



This is the results:




```
#######################################
# Program: water
# Rundate: Fri 18 Apr 2025 03:06:59
# Commandline: water
#    -asequence /corral4/main/objects/a/8/b/dataset_a8b599ae-5157-43f9-
ae0c-84067cb25e21.dat
#    -bsequence /corral4/main/objects/f/6/d/dataset_f6da880b-4583-4808-
8e39-197b1c8fc1f6.dat
#    -outfile /corral4/main/jobs/066/907/66907251/outputs/dataset_3860bc43-
86ff-46a9-881a-56ba3c05513d.dat
#    -gapopen 10.0
#    -gapextend 0.5
#    -brief yes
#    -aformat3 srs
```

```
#    -auto
# Align_format: srs
# Report_file:
/corral4/main/jobs/066/907/66907251/outputs/dataset_3860bc43-86ff-46a9-
881a-56ba3c05513d.dat
#######################################

#=======================================
#
# Aligned_sequences: 2
# 1: NM_000517.6_hg19
# 2: NM_000518.5_hg19
# Matrix: EBLOSUM62
# Gap_penalty: 10.0
# Extend_penalty: 0.5
#
# Length: 147
# Identity:      64/147 (43.5%)
# Similarity:    89/147 (60.5%)
# Gaps:           8/147 ( 5.4%)
# Score: 297.5
#
#
#=======================================

NM_000517.6_h      3 LSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHF-DLS-
50
NM_000518.5_h      4 LTPEEKSAVTALWGKV--NVDEVGGEALGRLLVVYPWTQRFFESFGDLST
51

NM_000517.6_h     51 ----HGSAQVKGHGKKVADALTNAVAHVDDMPNALSALSDLHAHKLRVDP
96
NM_000518.5_h     52 PDAVMGNPKVKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDP
101

NM_000517.6_h     97 VNFKLLSHCLLVTLAAHLPAEFTPAVHASLDKFLASVSTVLTSKYRZ    143
NM_000518.5_h    102 ENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYHZ    148


#=======================================
#
# Aligned_sequences: 2
# 1: NM_000517.6_hg19
# 2: NM_000518.5_equCab2
# Matrix: EBLOSUM62
# Gap_penalty: 10.0
# Extend_penalty: 0.5
#
# Length: 147
# Identity:      63/147 (42.9%)
# Similarity:    85/147 (57.8%)
# Gaps:           8/147 ( 5.4%)
# Score: 281.5
#
#
#=======================================

NM_000517.6_h      3 LSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHF-DLSH
51
NM_000518.5_e      4 LSGEEKAAVLALWDKVNEE--EVGGEALGRLLVVYPWTQRFFDSFGDLSN
51
```

```
NM_000517.6_h      52 -----GSAQVKGHGKKVADALTNAVAHVDDMPNALSALSDLHAHKLRVDP
96
NM_000518.5_e      52 PGAVMGNPKVKAHGKKVLHSFGEGVHHLDNLKGTFAALSELHCDKLHVDP
101

NM_000517.6_h      97 VNFKLLSHCLLVTLAAHLPAEFTPAVHASLDKFLASVSTVLTSKYRZ    143
NM_000518.5_e     102 ENFRLLGNVLVVVLARHFGKDFTPELQASYQKVVAGVANALAHKYHZ    148


#---------------------------------------
#---------------------------------------
```